

FREE SUPPLEMENTARY COPY ELECTROMAGNETIC FIELDS

(EC 3452)

For B.E. IV Semester ECE Branch

**As per the Latest Syllabus of Anna University, Chennai
(Regulations - 2021)**

**LATEST SOLVED ANNA UNIVERSITY
QUESTION PAPERS**

**APRIL / MAY 2023
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BY

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B.E / B.Tech DEGREE EXAMINATIONS, APRIL/MAY 2023
Fourth Semester, Electronics and Communication Engineering
EC 3452 – ELECTROMAGNETIC FIELDS

(Common to : Electronics and Telecommunication Engineering)

(Regulations 2021)

Time Three Hours

Maximum : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. *What is divergence of a vector field?*

Ans: Refer Question No. 11

Page No: 1.128

2. *Calculate the curl of gradient of the scalar field, $V = 3xy - yz$.*

$$\nabla \times \nabla V ; V = 3xy - yz$$

$$\nabla V = \frac{\partial}{\partial x} (3xy - yz) \bar{a}_x + \frac{\partial}{\partial y} (3xy - yz) \bar{a}_y + \frac{\partial}{\partial z} (3xy - yz) \bar{a}_z$$

$$\nabla V = 3y \bar{a}_x + (3x - z) \bar{a}_y - y \bar{a}_z$$

$$\nabla \times \nabla V = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3y & 3x - z & -y \end{vmatrix} = \bar{a}_x [-1 + 1] - \bar{a}_y [0 - 0] + \bar{a}_z [3 - 3]$$

$$\boxed{\nabla \times \nabla V = 0}$$

3. *Define Gauss's law.*

Ans: Refer Question No. 58

Page No: 1.142

4. *What is the significance of Laplacian Operator?*

Ans: Refer Question No. 3

Page No: 2.63

5. *Define Ampere's Law.*

Ans: Refer Question No. 4

Page No: 3.37

6. *What is the significance of magnetic vector potential?*

Ans: Refer Question No. 7

Page No: 3.38

UQ.2

Electromagnetic Fields

7. What is the displacement current?

The current flow through the capacitor is called as displacement current.

$$I_D = C \frac{dv}{dt}$$

$$I_D = \frac{\epsilon A}{d} \frac{dv}{dt} \quad V = E \cdot d$$

$$J_D = \frac{I_D}{A} = \frac{\epsilon \partial E}{\partial t} = \frac{\partial D}{\partial t}$$

8. What is the significance of Continuity Equation?**Continuity Equation**

The energy can neither be created (or) destroyed, it converts the energy from one form to another form with /without losses.

$$\nabla \cdot J = \frac{\partial \rho_v}{\partial t}$$

9. Define skin depth.

The 37% of its original wave, it is also called as depth of penetration

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

10. "X-rays can penetrate the human body, but light cannot". Justify.

X-rays have higher energy, so that they can penetrate through human body, where as light waves not have such a high energy..

PART B – (5 × 13 = 65 Marks)**11. (a) Explain different types of coordinate systems along with examples of their use.**

Ans: Refer Section No. 1.3

Page No: 1.6 – 1.10

[OR]

(b) Explain Gradient, Divergence and Curl in detail along with examples.

(3 + 5 + 5 = 13)

Ans: Refer Section No. 1.2.4

Page No: 1.5 – 1.6

12. (a) Determine the net electric flux leaving through closed surface defined by $\rho = 4$, $0 \leq z \leq 1$, if electric flux density is given by $D = \rho^2 \cos^2 \phi \bar{a}_\rho + z \sin \phi \bar{a}_\phi$ C/m².

$$D = \rho^2 \cos^2 \phi \bar{a}_\rho + z \sin \phi \bar{a}_\phi \text{ C/m}^2 \quad \rho = 4 ; 0 \leq z \leq 1$$

$$\psi = \iint D \cdot d\mathbf{s}$$

$$= \int_0^4 \int_0^{2\pi} \rho^2 \cos^2 \phi \bar{a}_\rho \cdot \rho d\phi dz \bar{a}_\rho + \iint z \sin \phi \bar{a}_\phi \cdot d\rho dz \bar{a}_\phi$$

$$= \int_0^1 \int_0^{2\pi} \phi^3 \cos^2 \phi d\phi dz + \int_0^1 \int_0^4 z \sin \phi d\rho dz$$

$$\psi = (4)^3 (1 - 0) \int_0^{2\pi} \left(\frac{1 + \cos \phi}{2} \right) d\phi + 0$$

$$= \frac{64}{2} \left[\phi + \frac{\sin 2\phi}{2} \right]_0^{2\pi} = 32 [2\pi + 0]$$

$$\boxed{\phi = 64\pi}$$

- (b) Derive the expression for Electric Field Intensity, E at a distance ; r due to a volume charge as sphere of radius, R carrying charge density, ρ_v C/m³. Consider

(i) $r < R$ (6.5)

(ii) $r > R$ (6.5)

$r > a$

$$d\phi = D \cdot d\mathbf{s} = D_r \bar{a}_r \cdot r^2 \sin \theta d\theta d\phi \bar{a}_r$$

$$d\phi = D_r r^2 \sin \theta d\theta d\phi$$

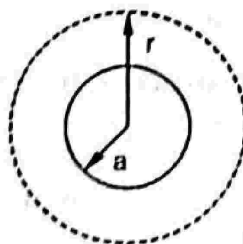


Fig. SQ.1.

UQ.4

By solving $D_r = \frac{Q}{4\pi r^2}$

$$E = \frac{D}{\epsilon_0} = \frac{Q}{4\pi \epsilon_0 r^2} \bar{a}_r$$

$r < a$

$$d\phi = D \cdot ds = \frac{Q}{4\pi r^2}$$

$$\bar{D} = \frac{\frac{4}{3} \pi r^3}{4\pi r^2}$$

$$\bar{D} = \frac{r}{3} \rho_v \bar{a}_r$$

$$\bar{E} = \frac{r}{3 \epsilon_0} \rho_v \bar{a}_r$$

[OR]

13. (a) Find the magnetic field intensity, H and magnetic flux density, B at the centre of the circular loop carries a current of 1 amperes in the clockwise direction at $z = 0$ plane. Assume the diameter of the circular loop is 2α .

Ans: Refer Section No. 3.3.2 Page No: 3.5 – 3.6

[OR]

- (b) For a current distribution in free space, magnetic vector potential is given by: $A = (2x^2z + yz)a_x + (xy^2 - xz^3)a_y - (6xyz - 2x^2y^2)A_z$ Wb/m.

Calculate:

- (i) Magnetic flux density, B and (5)
- (ii) the flux crossing through the surface described by $x = 1$, $0 < y < 2$, $0 < z < 2$. (8)

$$\bar{A} = (2x^2z + yz) \bar{a}_x + (xy^2 - xz^3) \bar{a}_y - (6xyz - 2x^2y^2) \bar{a}_z \text{ Wb/m}$$

$$\begin{aligned}
 \mathbf{B} = \nabla \times \mathbf{A} &= \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x^2z + yz & xy^2 - xz^3 & -(6xyz - 2x^2y^2) \end{vmatrix} \\
 &= \bar{a}_x \left[\frac{\partial}{\partial y} (6xyz - 2x^2y^2) - \frac{\partial}{\partial z} (xy^2 - xz^3) \right] - \\
 &\quad \bar{a}_y \left[\frac{\partial}{\partial x} (6xyz - 2x^2y^2) - \frac{\partial}{\partial z} (2xz^2 - yz) \right] + \\
 &\quad \bar{a}_z \left[\frac{\partial}{\partial x} (xy^2 - xz^3) - \frac{\partial}{\partial y} (2x^2z - yz) \right] \\
 &= -\bar{a}_x [-(6xz - 4x^2y) - (-3z^2x)] - \bar{a}_y [-(6yz - 4xy^2) \\
 &\quad - (2x^2 + y)] - \bar{a}_z [(y^2 - z^3) - z] \\
 \bar{\mathbf{B}} &= (3xz^2 + 4x^2y - 6xz)\bar{a}_x + (6yz - 4xy^2 + 2x^2 + y)\bar{a}_y + (z^3 - z - y^2)\bar{a}_z \\
 \psi &= \iint \mathbf{B} \cdot d\mathbf{s} \\
 &= \int_0^2 \int_0^2 (3xz^2 + 4x^2y - 6xz) dy dz + \int_0^2 \int_0^1 (6yz - 4xy^2 + 2x^2 + y) \\
 &\quad dx dz + \int_0^2 \int_0^1 (z^3 + z - y^2) dx dy \\
 &= \left(\left(\frac{3xyz^3}{3} + \frac{4x^2y^2z}{2} - \frac{6xyz^2}{2} \right)_0^2 \right)_0^2 + \\
 &\quad \left(\left(\frac{6xz^2}{2} - \frac{4x^2y^2z}{2} + \frac{2x^3z}{3} + xyz \right)_0^1 \right)_0^2 + \\
 &\quad \left(\left(xz^2y + xyz - \frac{xy^3}{3} \right)_0^1 \right)_0^2 \\
 \psi &= x(2)(2)^3 + 2(2)^2x^2(2) - 3x(2)(2)^2 + 3(1)(2)^2 - 2(1)y^2(2) \\
 &\quad + \frac{2}{3}(2) + 1(y)(2) + (1)(2)z^2 + 2z + \frac{(2)^3}{3} \\
 &= 16 + 16 - 24 + 12 - 4y^2 + \frac{4}{3} + 2y + 2z^2 + 2z + \frac{8}{3}
 \end{aligned}$$

UQ.6

$$\begin{aligned}
 &= 32 - 12 - 4y^2 + \frac{4}{3} + 2y + 2z^2 + 2z + \frac{8}{3} \\
 &= 32 - 12 - 16 - \frac{4}{3} + 4 + 8 + 4 + \frac{8}{3} \\
 &= 20 + \frac{4}{3} \\
 \psi &= \frac{64}{3}
 \end{aligned}$$

14. (a) Explain in detail Maxwell's equations in integral and differential form for time varying fields.

Ans: Refer Section No. 5.2

Page No: 5.9 – 5.14

[OR]

- (b) Explain the phenomenon of EM wave propagation in free space using Maxwell's Equations.

Ans: Refer Section No. 5.42

Page No: 5.22 – 5.24

15. (a) The electric field intensity of a uniform plane wave in an unknown medium is given by:

$$E(y, t) = 25 \sin(10^8 t - y) A_z \text{ V/m}$$

Calculate:

- (i) nature of the medium, (2)
- (ii) attenuation constant (α), (2)
- (iii) phase constant (β), (2)
- (iv) phase velocity (v_p) and (2)
- (v) derive the expression for magnetic field intensity, H (5)

$$E(y(t)) = 25 \sin(10^8 t - y) \bar{a}_z \text{ V/m}$$

$$\omega = 10^8 \quad \beta = 1$$

$$2\pi f = 10^8$$

$$f = \frac{10^8}{2\pi}$$

$$\lambda = \frac{2\pi}{\beta} = 2\pi$$

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UQ.7

$$V = f\lambda = \frac{10^8}{2\pi} \cdot 2\pi$$

$$V = 10^8 \text{ m/s}$$

$$V = \frac{C}{\sqrt{\mu_r \epsilon_r}}$$

$$\mu_r = 1$$

$$V = \frac{C}{\sqrt{\epsilon_r}}$$

$$\sqrt{\epsilon_r} = \frac{c}{v}$$

$$\epsilon_r = \left(\frac{c}{v}\right)^2 = \frac{3 \times 10^8}{10^8}$$

$$\epsilon_r = 3$$

$$\nabla \times \mathbf{E} = \frac{\partial \mathbf{B}}{\partial t}$$

$$\begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & 25 \sin(10^8 t - y) \end{vmatrix} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\bar{a}_x \left[\frac{\partial}{\partial y} (25 \sin(10^8 t - y)) - 0 \right] - \bar{a}_y [0] + \bar{a}_x [0] = -\frac{\partial \mathbf{B}}{\partial t}$$

$$-25 \cos(10^8 t - y) = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\mathbf{B} = \frac{25 \sin(10^8 t - y) \bar{a}_x}{10^8}$$

$$\mathbf{H} = \frac{25 \sin(10^8 t - y) \bar{a}_x}{4 \times 10^{-7} \times 10^8}$$

$$\mathbf{H} = 0.199 \sin(10^8 t - y) \bar{a}_x$$

[OR]

(b) In a certain lossy medium, an EM wave travels for a distance of 10 m where its amplitude decays to 1/e times of its initial value. if the phase

Electromagnetic Fields

UQ.8

shift for the same period is 60° , then calculate the propagation constant of the medium.

$$\text{Distance} = 10 \text{ m}$$

$$\text{Amplitude decays to } \frac{1}{e} = 37\%$$

$$\text{Phase shift for the same period} = 60^\circ$$

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

$$\gamma = \alpha + j\beta$$

$$\alpha = -\frac{h\left(\frac{1}{e}\right)}{10 \text{ m}} = -\frac{1}{10} \text{ m}^{-1}$$

$$60^\circ = \left(\frac{2\pi}{\lambda}\right) \times 10 \text{ m}$$

$$\lambda = \frac{2\pi}{60^\circ} \times 10 \text{ m}$$

$$\lambda = 5\pi \text{ m}$$

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi}{5\pi} = \frac{2}{5} \text{ m}^{-1}$$

$$\gamma = \alpha + j\beta = \left[-\frac{1}{10} + j\left(\frac{2}{5}\right)\right] \text{ m}^{-1}$$

$$\gamma = (-0.1 + 0.4j) \text{ m}^{-1}$$

PART C - (1 × 15 = 15 Marks)

16. (a) Three infinite sheets with charge density of 18 nC/m^2 , 9 nC/m^2 and -24 nC/m^2 are located at $x = 4$, $y = -3$, and $z = 0$ respectively. Find the electric field intensity at

(i) $(8, 0, 6)$ and

(7.5)

(ii) $(-2, -7, 1)$

(7.5)

$$\rho_{S1} = 18 \text{ nC/m}^2 \quad x = 4$$

$$\rho_{S2} = 9 \text{ nC/m}^2 \quad y = -3$$

$$\rho_{S3} = -24 \text{ nC/m}^2 \quad z = 0$$

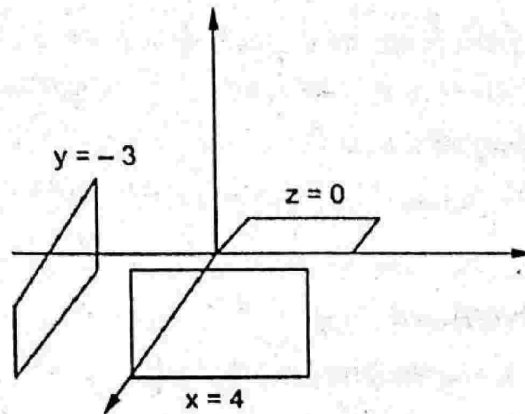


Fig. SQ.2.

(i) (8, 0, 6)

$$\begin{aligned}
 E &= E_1 + E_2 + E_3 \\
 &= \frac{\rho_{s1}}{2\epsilon_0} (4 \bar{a}_x) + \frac{\rho_{s2}}{2\epsilon_0} (3 \bar{a}_y) + \frac{\rho_{s3}}{2\epsilon_0} (6 \bar{a}_z) \\
 E_T &= \frac{18 \times 10^{-9}}{2\epsilon_0} \times 4 \bar{a}_x + \frac{9 \times 10^{-9}}{2\epsilon_0} (3 \bar{a}_y) - \frac{24 \times 10^{-9}}{2\epsilon_0} (6 \bar{a}_z) \\
 &= \frac{4.06 \times 10^{-9}}{10^{-12}} \bar{a}_x + \frac{1.525 \times 10^{-9}}{10^{-12}} \bar{a}_y - \frac{8.132 \times 10^{-9}}{10^{-12}} \bar{a}_z \\
 E_T &= (4.06 \bar{a}_x + 1.525 \bar{a}_y - 8.132 \bar{a}_z) \text{ kV/m}
 \end{aligned}$$

(ii) (-2, -7, 1)

$$\begin{aligned}
 E &= E_1 + E_2 + E_3 \\
 &= \frac{\rho_{s1}}{2\epsilon_0} (-6 \bar{a}_x) + \frac{\rho_{s2}}{2\epsilon_0} (-4 \bar{a}_y) + \frac{\rho_{s3}}{2\epsilon_0} (\bar{a}_z) \\
 &= \frac{-18 \times 10^{-9}}{2\epsilon_0} \times 6 \bar{a}_x + \frac{-9 \times 10^{-9}}{2\epsilon_0} \times 4 \bar{a}_y + \frac{-24 \times 10^{-9}}{2\epsilon_0} \bar{a}_z \\
 &= -\frac{54 \times 10^{-9}}{8.854 \times 10^{-12}} \bar{a}_x - \frac{18 \times 10^{-9}}{8.854 \times 10^{-12}} \bar{a}_y - \frac{12 \times 10^{-9}}{8.854 \times 10^{-12}} \bar{a}_z \\
 &= -6.0989 \times 10^3 \bar{a}_x - 2.033 \times 10^3 \bar{a}_y - 1.3553 \times 10^3 \bar{a}_z
 \end{aligned}$$

$$E = -6.0989 \bar{a}_x - 2.033 \times 10^3 \bar{a}_y - 1.3553 \times 10^3 \bar{a}_z \text{ K V/m}$$

[OR]

UQ.10

Electromagnetic Fields

(b) An EM wave travels from a free space to a dielectric medium with dielectric constant (ϵ_r) = 4 and it incidents normally on the interface. If the electric field of incident wave in free space is given by:

$$E_i = E_0 \cos(\omega t - \beta z) \hat{a}_y \text{ V/m, where } \omega = 3 \times 10^9 \pi \text{ and } \beta = 10 \pi$$

Then, Calculate the value of

(i) reflection coefficient (Γ_E), (3)

(ii) transmission coefficient (τ_E), (3)

(iii) the fraction of power transmitted into the dielectric medium and, (3)

(iv) derive the expression for electric field of the transmitted wave (E_T) (6)

$$\epsilon_r = 4, E_i = E_0 \cos(\omega t - \beta z) \hat{a}_y \text{ V/m}$$

$$\omega = 3 \times 10^9 \pi, \beta = 10 \pi$$

(i) Reflection coefficient (Γ) = $\frac{E_r}{E_i} = \frac{3}{5}$

(ii) Transmission coefficient $T = \frac{E_t}{E_i} = \frac{2}{5}$

(iii) Fraction of power transmitted = $(T)^2 = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$

(iv) $E_t = T \times E_i$
 $E_t = \frac{2}{5} \times E_0 \cos(\omega t - \beta z) \text{ V/m}$

B.E / B.Tech DEGREE EXAMINATIONS, NOVEMBER / DECEMBER 2023

Fourth Semester, Electronics and Communication Engineering

EC 3452 – ELECTROMAGNETIC FIELDS

(Common to Electronics and Telecommunication Engineering)

(Regulations 2021)

Time Three Hours.

Maximum : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. Define Scalar field.

The field which explains only the magnitude is called as scalar field. Eg: Mau.

2. State Divergence theorem.

Ans: Refer Question No. 36

Page No. 1.135

3. Define electric flux density.

The total number of electric lines per unit area is called as electric flux density

$$D = \frac{\phi}{A}$$

4. List two applications of Gauss's law.

Application of Gauss Law

(i) Used to determine the field intensity and flux density of infinite line charge

(ii) Used to determine the field intensity and flux density of infinite sheet charge

5. Differentiate static electric and magnetic field lines.

Electric Field Lines	Magnetic Field Line
Occur between the two opposite charges	Occur between the north pole to south pole
Electric flux density; $D = \frac{\phi}{A}$	Magnetic flux density; $B = \frac{\phi}{A}$

UQ.12

Electromagnetic Fields

6. What is a torque?

The rotational force is called as Torque.

$$T = R \times \bar{F} \text{ Unit is Nm}$$

7. State Faraday's law.

Ans: Refer Question No. 1

Page No. 5.55

8. What is potential difference?

Ans: Refer Question No. 62

Page No. 1.142

9. What is group velocity?

Ans: Refer Section No. 5.4

Page No. A.31

10. List four properties of good conductors.

Four properties of Good Conductor

(i) Allows the current through it

$$(ii) \alpha = \beta = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

(iii) $\epsilon = 0$

(iv) No resistivity

PART B - (5 × 13 = 65 Marks)

11. (a) Given, $A = (\sin 2\phi) a_\phi$ in cylindrical co-ordinates. Find curl of A at $(2, \pi/4, 0)$.(i) Spherical co-ordinates of A .(ii) Cartesian co-ordinate of B .

(7)

$$A = [\sin 2\phi] a_\phi \quad \nabla \times A = ? \text{ at } \left(2, \frac{\pi}{4}, 0 \right)$$

(6)

$$\nabla \times A = \frac{1}{r} \begin{vmatrix} a_r & r a_\phi & a_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_r & r A_\phi & A_z \end{vmatrix}$$

$$\begin{aligned}
 &= \frac{1}{r} \begin{vmatrix} a_r & r a_\phi & a_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ 0 & r \sin 2\phi & 0 \end{vmatrix} \\
 &= \frac{1}{r} \left[\bar{a}_r (0 - 0) - r \bar{a}_\phi (0 - 0) + \bar{a}_z \left[\frac{\partial}{\partial r} (r \sin 2\phi) \right] \right] \\
 &= \frac{1}{r} [0 - 0 + \bar{a}_z \sin 2\phi] \\
 \nabla \times \mathbf{A} &= \frac{\sin 2\phi}{r} \bar{a}_z \quad \left[\phi = \frac{\pi}{4}, r = 2, z = 0 \right] \\
 \nabla \times \mathbf{A} &= \frac{\sin 2\left(\frac{\pi}{4}\right)}{2} \bar{a}_z \\
 \nabla \times \mathbf{A} &= \frac{\sin\left(\frac{\pi}{2}\right)}{2} \bar{a}_z \\
 \nabla \times \mathbf{A} &= 0.5 \bar{a}_z
 \end{aligned}$$

[OR]

(b) Given the two points, $A(x = 2, y = 3, z = 1)$ and $B(r = 2, \theta = 20^\circ, \phi = 220^\circ)$ Find (13)

$A(x = 2, y = 3, z = 1)$ $B(r = 2, \theta = 20^\circ, \phi = 220^\circ)$

(i) Spherical Coordinate of A

$$x = 2 \quad r = \sqrt{x^2 + y^2 + z^2} = \sqrt{4 + 9 + 1} = \sqrt{14} = 3.74$$

$$y = 3 \quad \theta = \cos^{-1}\left(\frac{z}{r}\right) = \cos^{-1}\left(\frac{1}{3.74}\right) = 74.49$$

$$z = 1 \quad \phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{3}{2}\right) = \tan^{-1}(1.5) = 56.31$$

$$\mathbf{A}(r, \theta, \phi) = (3.74, 74.49, 56.31)$$

$$r = 2 \quad x = r \sin \theta \cos \phi \Rightarrow 2(\sin 20^\circ)(\cos 220^\circ) = -0.524$$

$$\theta = 20^\circ \quad y = r \sin \theta \sin \phi \Rightarrow 2(\sin 20^\circ)(\sin 220^\circ) = -0.434$$

$$\phi = 220^\circ \quad z = r \cos \theta \Rightarrow 2 \cos 20^\circ = 1.879$$

$$\mathbf{B}(x, y, z) = (-0.524, -0.434, 1.879)$$

UQ.14

Electromagnetic Fields

12. (a) A charge is distributed on x axis of Cartesian system having a line charge density of $3x^2 \mu\text{C/m}$. Find the total charge over the length of 10 m. (13)

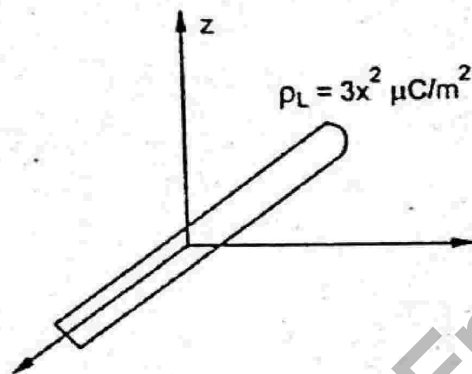


Fig. SQ.1.

$$\rho_L = \frac{dQ}{dL} \Rightarrow dQ = \rho_L dL$$

$$\begin{aligned} Q &= \int \rho_L dL \\ &= \int_0^{10} 3x^2 dx \\ &= 3 \left(\frac{x^3}{3} \right)_0^{10} \\ &= (x^3)_0^{10} \\ &= 1000 - 0 \end{aligned}$$

$$\boxed{Q = 1000 \text{ C}}$$

[OR]

- (b) If $V = x - y + xy + 2z$ Volt, Find

(i) E at $(2, 2, 1)$

(7)

- (ii) Energy stored in a cube of side 1 m centered at the origin.

(6)

If $V = (x - y + xy + 2z)$ Volt find at $(2, 2, 1)$

(i) $E = -\nabla V$

$$= - \left[\frac{\partial V}{\partial x} \bar{a}_x + \frac{\partial V}{\partial y} \bar{a}_y + \frac{\partial V}{\partial z} \bar{a}_z \right]$$

...(1)

$$\frac{\partial V}{\partial x} = \frac{\partial}{\partial x} (x - y + xy + 2z) = 1 - 0 + y + 0 = 1 + y \quad \dots(2)$$

$$\frac{\partial V}{\partial y} = \frac{\partial}{\partial y} (x - y + xy + 2z) = 0 - 1 + x + 0 = x - 1 \quad \dots(3)$$

$$\frac{\partial V}{\partial z} = \frac{\partial}{\partial z} (x - y + xy + 2z) = 0 - 0 + 0 + 2 = 2 \quad \dots(4)$$

Put (2), (3), (4) in (1)

$$E = - \left[(1 + y) \bar{a}_x + (x - 1) \bar{a}_y - 2 \bar{a}_z \right]$$

$$E = - (1 + y) \bar{a}_x + (x - 1) \bar{a}_y - 2 \bar{a}_z \quad (x = 2, y = 2, z = 1)$$

$$E = - (3) \bar{a}_x - \bar{a}_y - 2 \bar{a}_z \quad \text{V/m}$$

$$E = - 3 \bar{a}_x - \bar{a}_y - 2 \bar{a}_z \quad \text{V/m}$$

$$(ii) \text{ Energy stored} = \frac{1}{2} |E| |D| \quad |E| = \sqrt{9 + 1 + 4} = \sqrt{14}$$

$$= \frac{1}{2} \epsilon |E|^2$$

$$= \frac{1}{2} (8.854 \times 10^{-12})(14)$$

$$= 61.978 \times 10^{-12}$$

13. (a) Obtain the expression for magnetic field intensity on an axis of a circular ring. (13)

Ans: Refer Section No. 3.3.2

Page No. 3.5 to 3.6

[OR]

- (b) Find the magnetic field intensity at a point P, due to a finite straight conductor, carrying a current I. (13)

Ans: Refer Section No. 3.3.1

Page No. 3.3 to 3.5

14. (a) Derive wave equation, and explain the properties of uniform plane waves in free space. (13)

Ans: Refer Section No. 5.4.2

Page No. 5.22 to 5.24

Refer Section No. 5.4.5

Page No. 5.24 to 5.25

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Electromagnetic Fields

[OR]

- (b) Derive and explain, Maxwell's equations both in integral and point forms. (13)

Ans: Refer Section No.5.2.1

Page No.5.9 to 5.11

Refer Section No. 5.2.2

Page No. 5.12

Refer Section No. 5.2.3

Page No. 5.13

Refer Section No. 5.2.4

Page No. 5.13 to 5.14

15. (a) Derive pointing vector in integral and point form from Maxwell's equation. (13)

Ans: Refer Section No. 5.6.1

Page No. 5.30 to 5.33

[OR]

- (b) Explain the reflection of plane wave by conducting medium, under normal incidence. (13)

Ans: Refer Section No. 5.5

Page No. A. 35

PART C - (1 × 15 = 15 Marks)

16. (a) (i) A conductor 6m long, lies along Z direction with a current of 2A in $\vec{a}_z$ direction. Find the force experienced by conductor if $B = 0.08 \vec{a}_x(T)$ (8)

$I = 2A$ in $\vec{a}_z$, $B = 0.08 \vec{a}_x(T)$, length = 6 m.

$$F = I L \times B$$

$$= I B L \sin\theta \quad \theta = 90^\circ$$

$$= 2 \times 0.08 \times 10 \cdot \vec{a}_z$$

$$F = 1.6 \vec{a}_z$$

- (ii) Find the maximum torque on an 85 turns, rectangular coil with dimension (0.2 × 0.3) m, carrying current of 5 Amps in a field $B = 6.5T$. (7)

$N = 85$; $A = 0.2 \times 0.3$; $I = 5 A$; $B = 6.5 T$

$$T_{max} = NIAB = 85 \times 5 \times 0.06 \times 6.5$$

$$T_{max} = 165.75 \text{ Nm}$$

[OR]

(b) (i) Position of a point P is $(8, 120^\circ, 330^\circ)$. Express in cylindrical coordinate system. (7)

$$r = 8 \quad x = r \sin \theta \cos \phi \Rightarrow 8(\sin(120^\circ) (\cos 330^\circ)) = 6$$

$$\theta = 120^\circ \quad y = r \sin \theta \sin \phi \Rightarrow 8(\sin 120^\circ) (\sin 330^\circ) = -3.46$$

$$\phi = 330^\circ \quad z = r \cos \theta \Rightarrow 8 \cos(120^\circ) = -4$$

$$(x, y, z) = (6, -3.46, -4)$$

$$r = \sqrt{x^2 + y^2} = \sqrt{(6)^2 + (-3.46)^2}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-3.46}{6}\right)$$

$$Z = z = -4$$

(ii) Determine the divergence of the vector field

$$\bar{P} = x^2 y z \bar{a}_x + x z \bar{a}_z \quad (8)$$

$$\bar{P} = x^2 y z \bar{a}_x + x z \bar{a}_z$$

$$\nabla \times \bar{P} = \frac{\partial}{\partial x} (\bar{P}_y) + \frac{\partial}{\partial y} (\bar{P}_x) + \frac{\partial}{\partial z} (\bar{P}_z)$$

$$= \frac{\partial}{\partial x} (x^2 y z) + \frac{\partial}{\partial y} (0) + \frac{\partial}{\partial z} (x z)$$

$$= 2 x y z + 0 + x$$

$$\nabla \cdot \bar{P} = 2 x y z + x$$

UQ.18

Electromagnetic Fields

B.E/B.Tech DEGREE EXAMINATIONS, APRIL / MAY 2024

Fourth Semester

Electronics and Communication Engineering

EC 3452 – ELECTROMAGNETIC FIELDS

(Common to Electronics and Telecommunication Engineering)

(Regulations 2021)

Time: Three Hours

Maximum: 100 Marks

Answer All Questions

PART-A (10 × 2 = 20 Marks)

1. Transform the vector $\vec{Q} = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \vec{a}_x$ to cylindrical and spherical coordinates.

Given:

$$\vec{Q} = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \vec{a}_x - \frac{yz}{\sqrt{x^2 + y^2 + z^2}} \vec{a}_z$$

$$A_x = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}$$

$$A_y = 0$$

$$A_z = \frac{yz}{\sqrt{x^2 + y^2 + z^2}}$$

Cylindrical Coordinate

$$\vec{A} = A_p \vec{a}_p + A_\phi \vec{a}_\phi + A_z \vec{a}_z$$

$$A_p = A_x \cos \phi + A_y \sin \phi$$

$$= \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \cos \phi$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi$$

$$= \frac{-\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \sin \phi$$

$$A_z = A_z = \frac{yz}{\sqrt{x^2 + y^2 + z^2}}$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right) = \tan^{-1} \left(\frac{0}{\frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}} \right) = \tan^{-1}(0)$$

$$\phi = 0$$

$$A_\rho = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \cos(0)$$

$$A_\rho = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \quad [\cos(0) = 1]$$

$$A_\phi = \frac{-\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \sin(0)$$

$$A_\phi = 0$$

Cylindrical Coordinate (A_ρ , A_ϕ , A_z)

$$\left[\frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}}, 0, \frac{yz}{\sqrt{x^2 + y^2 + z^2}} \right]$$

Spherical Coordinate

$$\vec{A} = A_r \vec{a}_r + A_\theta \vec{a}_\theta + A_\phi \vec{a}_\phi$$

$$A_r = A_x \sin \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta$$

$$A_r = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \sin \theta \cos \phi + 0 + \frac{yz}{\sqrt{x^2 + y^2 + z^2}} \cos \theta$$

$$A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta$$

$$A_\theta = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \cos \theta \cos \phi - \frac{yz}{\sqrt{x^2 + y^2 + z^2}} \sin \theta$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi$$

$$A_\phi = \frac{-\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \sin \phi$$

$$\phi = 0$$

$$A_r = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \sin \theta + \frac{yz}{\sqrt{x^2 + y^2 + z^2}} \cos \theta$$

$$A_\theta = \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \cos \theta - \frac{yz}{\sqrt{x^2 + y^2 + z^2}} \sin \theta$$

$$A_\phi = 0$$

UQ.20*Electromagnetic Fields*

Spherical coordinate is given by (A_r, A_θ, A_ϕ)

$$\left[\frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \sin \theta + \frac{yz}{\sqrt{x^2 + y^2 + z^2}} \cos \theta, \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2 + z^2}} \cos \theta - \frac{yz}{\sqrt{x^2 + y^2 + z^2}} \sin \theta, 0 \right]$$

2. Define the divergence theorem and stokes theorem.

Divergence Theorem

The integral of the normal component of any vector field over a closed surface is equal to the integral of the divergence of this vector field over a volume enclosed by the closed surface.

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \iiint_V \nabla \cdot \mathbf{D} dv$$

Stoke's Theorem

The line integral of a vector around a closed path is equal to the surface integral of the normal component of its curl over any surface bounded by the path.

$$\oint \mathbf{H} \cdot d\mathbf{l} = \iint \nabla \times \mathbf{H} \cdot d\mathbf{s}$$

- 3. Two point charges with $q_1 = 2 \times 10^{-5} \text{ C}$ and $q_2 = -4 \times 10^{-5} \text{ C}$ are located in free space at $(1, 3, -1)$ and $(-3, 1, -2)$ respectively, in a Cartesian Coordinate system. Find the electric field $\bar{E}$ at $(3, 1, -2)$ and the force of an $8 \times 10^{-5} \text{ C}$ charge located at that point. All distances are in metres.**

Given:

$$q_1 = 2 \times 10^{-5} \text{ C at } (1, 3, -1)$$

$$q_2 = -4 \times 10^{-5} \text{ C at } (-3, 1, -2)$$

Force of charge $q = 8 \times 10^{-5} \text{ C}$ at $(3, 1, -2)$

Distance from q and q_1 from

$$\begin{aligned} r_1 &= \sqrt{(3-1)^2 + (1-3)^2 + (-2+1)^2} \\ &= \sqrt{4+4+1} = \sqrt{9} \\ r_1 &= 3 \text{ m} \end{aligned}$$

Distance from q and q_2

$$\begin{aligned} r_2 &= \sqrt{(3+3)^2 + (1-1)^2 + (-2+2)^2} \\ &= \sqrt{36} \\ r_2 &= 6 \text{ m} \end{aligned}$$

$$\text{Electric field } E_1 = \frac{K q_1}{r_1^2} = \frac{9 \times 10^9 \times 2 \times 10^{-5}}{(3)^2}$$

$$E_1 = 2 \times 10^4 \text{ N/C}$$

$$\text{Electric field } E_2 = \frac{K q_2}{r_2^2} = \frac{9 \times 10^9 \times -4 \times 10^{-5}}{(6)^2}$$

$$E_2 = -1 \times 10^4 \text{ N/C}$$

E_1 is positive

E_2 is negative

$$E_{\text{total}} = E_1 \pm E_2$$

$$= 2 \times 10^4 \text{ N/C} \pm 1 \times 10^4 \text{ N/C}$$

$$E_{\text{total}} = 1 \times 10^4 \text{ N/C if subtract}$$

$$E_{\text{total}} = 3 \times 10^4 \text{ N/C if add}$$

$$F = q \times E_{\text{total}}$$

$$= 8 \times 10^{-5} \times E_{\text{total}}$$

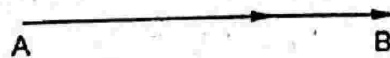
$$F = 8 \times 10^{-5} \times 1 \times 10^4 = 0.8 \text{ N}$$

$$F = 8 \times 10^{-5} \times 3 \times 10^4 = 2.4 \text{ N}$$

4. **Justify the statement. "Total electrostatic field of any closed loop is zero".**

The total electrostatic field in a closed loop is zero because electrostatic fields are conservative. This means that the work done in moving a charge around a closed loop is always zero, resulting in no net electric field. According to Gauss's Law, if no net charge is enclosed by the loop, the electric flux through it is zero, which also implies that the total electrostatic field around the loop cancels out.

5. **Predict the direction of the magnetic field, when current is passing from point A to B (y direction)**



Using the Right-Hand Thumb Rule, if the current flows from point A to B, point your thumb in that direction. The direction your fingers curl around the wire shows the magnetic field's direction, which forms concentric circles around the wire.

6. **A current loop experience a torque in a magnetic field, Justify that the force exerted on the whole loop is zero.**

A current loop in a magnetic field experiences equal and opposite forces on its opposite sides. These forces cancel each other out, resulting in a net force of zero on the entire loop. However, the unequal distribution of forces generates a torque, causing the loop to rotate without translating in space. Thus, while the loop experiences a torque, the net force acting on it is zero.

7. **Recall the four Maxwell equations for time-varying fields.**

$$\nabla \times \mathbf{H} = \mathbf{J} + j\omega \mathbf{D} = (\sigma + j\omega \epsilon) \mathbf{E}$$

Q.22

Electromagnetic Fields

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\frac{\partial \mu_0 \mathbf{H}}{\partial t}$$

$$\nabla \cdot \mathbf{D} = \rho$$

$$\nabla \cdot \mathbf{B} = 0$$

8. **How to overcome the inconsistency of Ampere's law in the time-varying field?**

To overcome the inconsistency of Ampere's Law in time-varying fields, Maxwell introduced the displacement current term. The modified Ampere's Law is expressed as:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \left[I_{\text{enc}} + \epsilon_0 \frac{d\phi_E}{dt} \right]$$

Here, the displacement current term accounts for changing electric fields, ensuring that the law is consistent with other Maxwell's equations. This unification allows for the proper description of electromagnetic phenomena, even in dynamic situations.

9. **Rearrange the Poynting vector, when the wave is propagated through a pure dielectric medium.**

In a pure dielectric medium, the Poynting vector, which represents the energy flux of an electromagnetic wave, is given by:

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{H}$$

In this medium, we can express the magnetic field as, where η is the intrinsic impedance of the medium. Substituting this in the Poynting vector becomes:

$$\mathbf{S} = \frac{1}{\mu_0 \eta} \mathbf{E} \times \mathbf{E}$$

Since, this implies that the Poynting vector's magnitude is typically considered as:

$$S = \frac{1}{\eta} |\mathbf{E}|^2$$

This indicates the energy flow and intensity in the medium.

10. **A plane wave propagating through a dielectric medium with $\epsilon_r = 8$, $\mu_r = 2$ and $E = 0.5 e^{-z/\beta} \sin(10^8 t - \beta z) \mathbf{a}_y$ V/m. Find the phase constant and skin depth.**

Given:

$$\mathbf{E} = 0.5 e^{-z/\beta} \sin(10^8 t - \beta z) \mathbf{a}_y \text{ V/m}$$

$$\epsilon_r = 8, \mu_r = 2$$

$$\omega = 10^8 = 2\pi f$$

$$f = \frac{10^8}{2\pi} = 15.91 \text{ MHz}$$

$$\beta = \frac{2\pi}{\lambda}$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{15.91 \times 10^6} = 18.86$$

$$\beta = \frac{2\pi}{18.86} = 0.33$$

$$\begin{aligned} \text{Skin depth } \delta &= \frac{1}{\sqrt{\pi f \mu \sigma}} \\ &= \frac{1}{\sqrt{\pi \times 15.91 \times 10^6 \times 4\pi \times 10^{-7} \times \sigma}} \\ \delta &= 0.126 \sigma \end{aligned}$$

PART B - (5 × 13 = 65 Marks)

11. (a) (i) *Analyze the Gradient of scalar and divergence and curl of the vector.* (6)

Ans: Refer Section No. 1.24

Page No. 1.5

- (ii) Find the gradient of the scalar fields $U = x^2 y + xyz$ and $U = e^{-x} \sin 2x \cos y$. (7)

Given:

(i)

$$U = x^2 y + x y z$$

$$\nabla U = \frac{\partial u}{\partial x} \vec{a}_x + \frac{\partial u}{\partial y} \vec{a}_y + \frac{\partial u}{\partial z} \vec{a}_z$$

$$= \frac{\partial}{\partial x} (x^2 y + x y z) \vec{a}_x + \frac{\partial}{\partial y} (x^2 y + x y z) \vec{a}_y + \frac{\partial}{\partial z} (x^2 y + x y z) \vec{a}_z$$

$$\nabla U = (2x + yz) \vec{a}_x + (x^2 + xz) \vec{a}_y + (xy) \vec{a}_z$$

(ii)

$$U = e^{-x} \sin 2x \cos y$$

$$\nabla U = \frac{\partial u}{\partial x} \vec{a}_x + \frac{\partial u}{\partial y} \vec{a}_y + \frac{\partial u}{\partial z} \vec{a}_z$$

$$= \frac{\partial}{\partial x} (e^{-x} \sin 2x \cos y) \vec{a}_x + \frac{\partial}{\partial y} (e^{-x} \sin 2x \cos y) \vec{a}_y + \frac{\partial}{\partial z} (e^{-x} \sin 2x \cos y)$$

$$\nabla U = (2 \cos 2x e^{-x} \cos y) \vec{a}_x - (e^{-x} \sin 2x \sin y) \vec{a}_y - (e^{-x} \sin 2x \cos y) \vec{a}_z$$

[OR]

UQ.24

Electromagnetic Fields

(b) (i) Convert points $P(1,3,5)$ and $T(0,-4,3)$ from Cartesian to cylindrical and Spherical coordinates. (5)

Given

(i) $P(1,3,5)$ (ii) $T(0,-4,3)$

Cylindrical Coordinates

$$x = 1; y = 3; z = 5$$

$$\begin{aligned} \rho &= \sqrt{x^2 + y^2} \\ &= \sqrt{1^2 + 3^2} = \sqrt{1+9} = \sqrt{10} \end{aligned}$$

$$\rho = 3.16$$

$$\begin{aligned} \phi &= \tan^{-1}\left(\frac{y}{x}\right) \\ &= \tan^{-1}\left(\frac{3}{1}\right) \end{aligned}$$

$$\phi = 71.56$$

$$z = z$$

$$z = 5$$

Spherical Coordinate

$$\begin{aligned} r &= \sqrt{x^2 + y^2 + z^2} \\ &= \sqrt{1^2 + 3^2 + 5^2} = \sqrt{1+9+25} = \sqrt{35} \end{aligned}$$

$$r = 5.92$$

$$\theta = \cos^{-1}\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right)$$

$$\theta = \cos^{-1}\left(\frac{5}{5.92}\right)$$

$$\theta = 32.37$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$= \tan^{-1}\left(\frac{3}{1}\right)$$

$$\phi = 71.56$$

(ii) $T(0, -4, 3)$

Cylindrical Coordinate

$$\begin{aligned} \rho &= \sqrt{x^2 + y^2} \\ &= \sqrt{0^2 + (-4)^2} = \sqrt{16} \end{aligned}$$

Solved Anna University Question Paper

UQ.25

$$P = 4$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

$$= \tan^{-1} \left(\frac{-4}{0} \right)$$

$$\phi = \infty$$

$$z = z$$

$$z = 3$$

Spherical Coordinate

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$= \sqrt{0^2 + (-4)^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25}$$

$$r = 5$$

$$\theta = \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right)$$

$$= \cos^{-1} \left(\frac{3}{5} \right)$$

$$\theta = 53.13^\circ$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

$$\phi = \tan^{-1} \left(\frac{-4}{0} \right)$$

$$\phi = \infty$$

- (ii) Compute the divergence and curl of the vector field $A = yz \bar{a}_x + 4xy \bar{a}_y + y \bar{a}_z$ and evaluate it at the point $(1, -2, 3)$. (7)

Given:

$$A = yz \bar{a}_x + 4xy \bar{a}_y + y \bar{a}_z$$

Divergence of vector field

$$\nabla A = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$= \frac{\partial}{\partial x} (yz) + \frac{\partial}{\partial y} (4xy) + \frac{\partial}{\partial z} (y)$$

$$= 0 + 4x + 0$$

$$\nabla A = 4x$$

Q.28

Electromagnetic Fields

Points (1, -2, 3)

$$x=1, y=-2, z=3$$

$$\nabla A = 4(1)$$

$$\nabla A = 4$$

Curl of the vector field

$$\begin{aligned}\nabla \times A &= \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & 4xy & y \end{vmatrix} \\ &= \vec{a}_x \left[\frac{\partial}{\partial y} (y) - \frac{\partial}{\partial z} (4xy) \right] - \vec{a}_y \left[\frac{\partial}{\partial x} (y) - \frac{\partial}{\partial z} (yz) \right] \\ &\quad + \vec{a}_z \left[\frac{\partial}{\partial x} (4xy) - \frac{\partial}{\partial y} (yz) \right] \\ &= \vec{a}_x [1 - 0] - \vec{a}_y [0 - 1] + \vec{a}_z [4y - z] \\ &= \vec{a}_x + \vec{a}_y + \vec{a}_z [4y - z]\end{aligned}$$

At point (1, -2, 3)

$$x=1, y=-2, z=3$$

$$\nabla \times A = \vec{a}_x + \vec{a}_y + \vec{a}_z [4(-2) - 3]$$

$$\nabla \times A = \vec{a}_x + \vec{a}_y - 11 \vec{a}_z$$

12. (a) (i) If $D = (2y^2 + z) \vec{a}_x + 4xy \vec{a}_y + x \vec{a}_z$ C/m², find (7)

(1) The volume charge density is at (-1, 0, 3)

(2) The flux through the cube is defined by $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$.

(3) The cube encloses the total charge.

(i) Given

$$D = (2y^2 + z) \vec{a}_x + 4xy \vec{a}_y + x \vec{a}_z$$

1. The volume charge density is at (-1, 0, 3)

$$\nabla \cdot D = \rho$$

$$\nabla \cdot D = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

Solved Anna University Question Paper

UQ.27

$$= \frac{\partial}{\partial x} (2y^2 + z) + \frac{\partial}{\partial y} (4xy) + \frac{\partial}{\partial z} (x)$$

$$= 0 + 4x + 0$$

$$\nabla D = 4x$$

$$x = -1, y = 0, z = 3$$

$$\nabla D = 4(-1)$$

$$\rho_v = \nabla \cdot D = -4$$

Volume charge density

$$\rho_v = -4 \text{ Cm}^3$$

2. Flux through the cube

$$\phi = \int_S D \cdot dx dy dz$$

Face at $x = 1$

$$D = (2y^2 + z) dx$$

$$\phi_{x=1} = \int_0^1 \int_0^1 (2y^2 + z) dz dy$$

$$= \int_0^1 \left[2y^2 + \frac{z^2}{2} \right]_0^1 dy$$

$$= \int_0^1 \left[2y^2 + \frac{1}{2} \right] dy$$

$$= \left[\frac{2y^3}{3} + \frac{1}{2} \right]_0^1$$

$$= \left[\frac{2}{3} + \frac{1}{2} \right]$$

$$= \frac{4+3}{6} = \frac{7}{6}$$

Face at $x = 0$

$$\phi_{x=0} = 0$$

Face at $y = 1$

$$D = 4x dy$$

UQ.28*Electromagnetic Fields*

$$\begin{aligned}\phi_{y=1} &= \int_0^1 \int_0^1 4x \, dz \, dx \\ &= \int_0^1 4x \, dx = [2x^2]_0^1 = 2\end{aligned}$$

Face $y = 0$

$$\phi_{y=0} = 0$$

Face $z = 1$

$$D = \rho \, d.$$

$$\begin{aligned}\phi_{z=1} &= \int_0^1 \int_0^1 x \, dx \, dy = \int_0^1 x \, dx = \left[\frac{x^2}{2} \right] \\ \phi_{z=1} &= \frac{1}{2}\end{aligned}$$

Face at $z = 0$

$$\phi_{z=0} = 0$$

Total Flux

$$\phi_{total} = \frac{7}{6} + 0 + 2 + 0 + \frac{1}{2} + 0 = \frac{19}{6} \, \text{C}$$

Total flux through the cube is $\frac{19}{6} \, \text{C}$ **3. Cube enclosing total charge**

$$\phi = Q_{enclosed}$$

The total enclosed charge is $\frac{19}{6} \, \text{C}$ **(ii) Rearrange Gauss's law and develop Laplace's and Poisson's equations. (6)****Ans:** Refer Section No. 1.10 and 2.8 Page No. 1.28 and 2.20**[OR]**

(b) (i) Interpret the Electric Flux density for a uniformly charged sphere of radius 'a'. Construct a Gaussian surface for the case of $r \geq a$, and $r \leq a$ separately. (6)

Given:

$$\text{Radius} = a$$

Solution:

Gauss's law is given by

$$\oint \mathbf{D} \cdot d\mathbf{A} = Q$$

Case (i) $r \geq a$

Outside the sphere

The electric flux density D is radially symmetric and the same magnitude at all points on the surface of the Gaussian sphere.

From Gauss law

$$\oint \mathbf{D} \cdot d\mathbf{A} = Q$$

Charge is distributed over the spheres total charge enclosed by the Gaussian surface.

The area of the Gaussian surface is $4\pi r^2$.

$$D \cdot 4\pi r^2 = Q_{enc}$$

$$D = \frac{Q_{enc}}{4\pi r^2}$$

Case (ii) $r \leq a$

Inside the sphere

In this case Gaussian surface is inside the uniformly charged sphere.

The charge enclosed within the Gaussian surface is proportional to the volume enclosed by the surface.

$$Q_{enc} = Q \cdot \frac{V_{enclosed}}{V_{total}}$$

$$Q_{enc} = Q \cdot \frac{r^3}{a^3}$$

Now applying Gauss's law

$$D \cdot 4\pi r^2 = Q \cdot \frac{r^3}{a^3}$$

$$D = \frac{Q}{4\pi r^2} \times \frac{r^3}{a^3}$$

$$D = \frac{Q}{4\pi a^3} r$$

(ii) A parallel plate capacitor has a plate area of 200 m^2 and a plate separation of 3 cm . The charge density is with air dielectric. Determine

(1) The capacitance of the capacitor (4)

(2) The voltage between the plates (3)

UQ.30*Electromagnetic Fields***Given:**

$$A = 200 \text{ m}^2$$

$$d = 3 \text{ cm} = 3 \times 10^{-2} \text{ m}$$

$$\epsilon_r = 1$$

1. Capacitance of the capacitor

$$C = \frac{\epsilon_0 A}{d} = \frac{8.854 \times 10^{-12} \times 200}{3 \times 10^{-2}}$$

$$C = 5.903 \times 10^{-8}$$

$$C = 59.03 \text{ nF}$$

2. Voltage between the plates

$$V = \frac{Q}{C}$$

The total charge is given by

$$Q = \sigma A$$

Electric field is given by

$$E = \frac{\sigma}{\epsilon_0}$$

$$E = \frac{V}{d}$$

Combining two equation

$$\frac{V}{d} = \frac{\sigma}{\epsilon_0}$$

$$V = \frac{\sigma d}{\epsilon_0} = \frac{\sigma \times 3 \times 10^{-2}}{8.854 \times 10^{-12}}$$

$$V = 3388.30 \sigma \text{ volts}$$

13. (a) *Prove that total magnetic field intensity (H) outside of the outer coaxial conductor is zero for infinitely long coaxial transmission line using Amperes law. Determine H at each Amperian path.*

Ans: Refer Section No. 3.4.3

Page No. 3.14

[OR]

- (b) *Determine the Magnetic field and current distributions for the following three conditions*

- (i) *Infinite line current along the z-axis*

(4)

(ii) Infinite sheet of current (4)

(iii) Infinitely long coaxial transmission line (5)

Ans: Refer Section No. 3.4.1, 3.4.2 and 3.4.3

Page No. 3.12, 3.13 and 3.14

14. (a) (i) A thin ring of radius 5 cm is placed on plane $z = 1$ cm so that its center is at $(0, 0, 1)$ cm. If the ring carries 50 mA $\vec{a}_\phi$, find H at $(0, 0, -1)$ cm and $(0, 0, 10)$ cm. (7)

(i) Given

$$\text{Radius} = 5 \text{ cm} = 0.05 \text{ m}$$

$$Z = 1 \text{ cm}$$

$$I = 50 \text{ mA} = 0.05 \text{ A}$$

Magnetic field

$$H_z = \frac{I r^2}{2(r^2 + z^2)^{3/2}}$$

H at $(0, 0, -1)$ and $(0, 0, 10)$ $z = 1$ cm the total distance is

$$z = 1 - (-1) = 2 \text{ cm} = 0.02 \text{ m}$$

$$z = 10 - 1 = 9 \text{ cm} = 0.09 \text{ m}$$

For $(0, 0, -1)$; $z = 0.02$ m

$$H_z = \frac{0.05 \times (0.05)^2}{2[(0.05)^2 + (0.02)^2]^{3/2}}$$

$$H_z = \frac{1.25 \times 10^{-4}}{2(2.9 \times 10^{-3})^{3/2}}$$

$$H_z = \frac{1.25 \times 10^{-4}}{2 \times 1.56 \times 10^{-4}}$$

$$\boxed{H_z = 0.4 \text{ A/m}}$$

For $(0, 0, 10)$ cm; $z = 0.09$ m

$$H_z = \frac{0.05 \times (0.05)^2}{2[(0.05)^2 + (0.09)^2]^{3/2}}$$

$$= \frac{1.25 \times 10^{-4}}{2(0.0106)^{3/2}} = \frac{1.25 \times 10^{-4}}{2 \times 1.09 \times 10^{-3}}$$

$$\boxed{H_z = 0.0573 \text{ A/m}}$$

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Electromagnetic Fields

(ii) Prove the Maxwell's equations are related to time-varying magnetic fields..(6)

Ans: Refer Section No. 5.2.5

Page No. 5.15

[OR]

(b) (i) Reconstruct Ampere's circuit law for time-varying situations to satisfy Faraday's law. (7)

Ans: Refer Section No. 4.1

Page No. Appendix A20

(ii) Derive the Helmholtz's wave equations for both E and H fields. (6)

If the medium is vacuum or free space in which

$$\mu = \mu_0, \epsilon = \epsilon_0 \text{ and } \sigma = 0$$

The wave equation for free space is given by

$$\nabla^2 E = \mu \epsilon \frac{\partial^2 E}{\partial t^2}$$

$$\nabla^2 E = \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}$$

The phasor value of E is

$$E(x, t) = \text{Re} [E(x) e^{j\omega t}]$$

Applying to the wave equation

$$\nabla^2 \text{Re}[E e^{j\omega t}] = \mu_0 \epsilon_0 \frac{\partial^2}{\partial t^2} \text{Re}[E e^{j\omega t}]$$

$$\nabla^2 \text{Re}[E e^{j\omega t}] = \mu_0 \epsilon_0 \text{Re}[-\omega^2 e^{j\omega t}]$$

$$\text{Re}[(\nabla^2 E + \mu_0 \epsilon_0 \omega^2 E) e^{j\omega t}] = 0$$

$$\nabla^2 E + \mu_0 \epsilon_0 \omega^2 E = 0$$

This is the wave equation for lossless medium in phasor form and it is called vector Helmholtz equation.

$$\nabla^2 E + \gamma^2 E = 0$$

$$\nabla^2 E = -\gamma^2 E$$

$$\gamma^2 = -\omega^2 \mu_0 \epsilon_0$$

$$\gamma = j\omega \sqrt{\mu_0 \epsilon_0}$$

$$\gamma = \alpha + j\beta$$

Comparing real and imaginary parts

$$\text{Attenuation constant } \alpha = 0$$

$$\text{Phase constant } \beta = \omega \sqrt{\mu_0 \epsilon_0} = \omega/v$$

where

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$$

15. (a) Conclude that the tangential components of $\vec{H}$ are discontinuous across the boundary, and the normal components of $\vec{H}$ are continuous across the dielectric-dielectric boundary medium. Besides, determine $\vec{H}$'s tangential and normal components across the dielectric-conductor boundary medium.

Ans: Refer Section No. 4.7

Page No. 4.11

[OR]

- (b) (i) A uniform plane wave propagating in a lossless medium has $\vec{E} = 2 \sin [10^8 t - \beta z] \vec{a}_x \text{ V/m}$. If $\epsilon_r = 1$, $\mu_r = 2$ and $\sigma = -3 \text{ V/m}$, characterize the medium. Compute the $\eta\beta$ and H . (7)

(i) Given

$$\epsilon_r = 1; \mu_r = 2; \sigma = -3 \text{ V/m}$$

$$\vec{E} = 2 \sin [10^8 t - \beta z] \vec{a}_x \text{ V/m}$$

$$E = A \sin (\omega t - \beta z) \vec{a}_x$$

$$\omega = 10^8$$

$$2\pi f = 10^8$$

$$f = \frac{10^8}{2\pi}$$

$$f = 15.91 \text{ MHz}$$

(i) η

$$\text{Intrinsic impedance } \eta = \sqrt{\frac{\mu_0 \mu_r}{\epsilon_0 \epsilon_r}} = \sqrt{\frac{4\pi \times 10^{-7} \times 2}{8.854 \times 10^{-12} \times 1}}$$

$$\eta = 532.78 \Omega$$

(ii) β

$$V = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 2 \times 8.854 \times 10^{-12} \times 1}}$$

$$V = 211.99 \times 10^6 = 212 \times 10^6 \text{ m/s}$$

$$\beta = \frac{\omega}{V} = \frac{10^8}{212 \times 10^6}$$

$$\beta = 0.472 \text{ rad/m}$$

UQ.34*Electromagnetic Fields*(iii) **H**

$$\eta = \frac{E}{H}$$

$$H = \frac{E}{\eta} = \frac{2 \sin(10^8 t - \beta z) \vec{a}_z}{532.78}$$

$$H = 0.00375 \sin(10^8 t - \beta z) \vec{a}_z$$

(ii) If the wave encounters a perfectly conducting plate normal to the z-axis at $z = 0$, find the reflected wave E_r and H_r . (6)

Ans: Refer Section No. A5.5

Page No. A.33

PART C - (1 × 15 = 15 Marks)

16. (a) Develop the transmission and reflection coefficient expression when the incident wave from medium 2 propagates to medium 1 in normal incidence. Assume Medium 2 is air and medium 1 is Polyethylene with $\epsilon_r = 2.25$; $\mu_r = 1$. (15)

Given

Medium 1 → Polyethylene

$$\epsilon_r = 2.25$$

$$\mu_r = 1$$

Medium 2 → Air

$$\epsilon_r = \mu_r = 1$$

Step 1

$$\text{Reflection coefficient} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\eta_1 = \sqrt{\frac{\mu_1}{\epsilon_1}}$$

$$\eta_2 = \sqrt{\frac{\mu_2}{\epsilon_2}}$$

$$\eta_1 = \sqrt{\frac{1}{2.25}} = 0.66$$

$$\eta_2 = \sqrt{\frac{1}{1}} = 1$$

$$\text{Reflection coefficient} = \frac{1 - 0.66}{1 + 0.66} = \frac{0.34}{1.66}$$

$$\text{Reflection coefficient} = 0.205$$

Solved Anna University Question Paper

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Step 2:

$$\begin{aligned}\text{Transmission Coefficient } T &= \frac{2 \eta_2}{\eta_2 + \eta_1} \\ &= \frac{2 \times 1}{0.66 + 1} = \frac{2}{1.66} = 1.205\end{aligned}$$

Step 3: Energy Conservation

$$(\text{Reflection coefficient})^2 + (\text{Transmission coefficient})^2 = 1$$

$$(0.205)^2 + (1.205)^2 = 1$$

$$0.042 + 1.452 = 1$$

$$1.49405 = 1$$

This shows slightly rounding errors, but the result will ~~approximately satisfy the~~ conservation of energy.

[OR]

(b) Discuss the variation of flux with time in the following three cases:

(i) A stationary loop in a time-varying magnetic field (~~Transverse emf~~) (5)

Ans: Refer Question No. 5

Page No. 5.58

(ii) A time-varying loop in a static magnetic field (~~Motional emf~~) (5)

Ans: Refer Question No. 5

Page No. 5.58

(iii) A time-varying loop in a time-varying magnetic field. (5)

Ans: Refer Section No. 5.1 in (III)

Page No. 5.4

Reg. No. :

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Question Paper Code : 40984

B.E./B.Tech. DEGREE EXAMINATIONS, NOVEMBER/DECEMBER 2024.

Fourth Semester

Electronics and Communication Engineering

EC 3452 – ELECTROMAGNETIC FIELDS

(Common to : Electronics and Telecommunication Engineering)

(Regulations 2021)

Time : Three hours

Maximum : 100 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. Transform the cartesian coordinates (1, 2, 3) into spherical coordinates.
2. State Helmholtz's theorem.
3. Find the value of k , if the electric flux density in a charge free region is given by $D = 5x\bar{a}_x + 10y\bar{a}_y - kz^2\bar{a}_z$.
4. Determine the energy stored in parallel plate capacitor of 10cm by 10cm has a separation of 1 cm and a voltage difference of 10V.
5. State Biot Savart's law.
6. Distinguish between magnetic scalar potential and magnetic vector potential.
7. In a material for which $\sigma = 5$ s/m, $\epsilon_r = 1$ and $E = 100 \sin 10^6 t$ v/m. Find its displacement current density.
8. State Guass's law for magnetic field and express in integral form.
9. Define group velocity. Write its expression.
10. Calculate the characteristic impedance of free space.

PART B — (5 × 13 = 65 marks)

11. (a) State and verify Divergence theorem for its vector $A = 4x\bar{a}_x - 2y^2\bar{a}_y + z^2\bar{a}_z$ taken over the cube bounded by $x=0$, $x=1$, $y=0$ and $y=1$.

Or

- (b) State and verify Stoke's theorem. Give its application.

12. (a) Derive

(i) Laplace equation (7)

(ii) Equation of continuity. (6)

Or

- (b) Calculate the capacitance of a parallel plate capacitor with two dielectrics ($\epsilon_{r1} = 2$ of thickness 1 mm, $\epsilon_{r2} = 1$ of thickness 2 mm) kept parallel to the plates of area of 50 cm². What is the voltage gradient in each dielectric if 100 V is applied across its plates.

13. (a) Derive an expression for the magnetic field intensity on the axis of solenoid.

Or

- (b) Derive expression for inductance of

(i) solenoid (7)

(ii) toroid (6)

14. (a) State and derive Maxwell's equation in integral form.

Or

- (b) Derive wave equation for conducting medium.

15. (a) Obtain the expression for reflection coefficient and transmission coefficient when a wave incident normally at a dielectric boundary.

Or

- (b) State and prove Poynting theorem.

PART C — (1 × 15 = 15 marks)

16. (a) Show that the displacement current through the parallel plate capacitor of capacitance C is equal to the conduction current I if a supply voltage $V = V_m \sin \omega t$ is applied across the capacitor.

Or

- (b) A uniform plane wave is travelling at a velocity of 2.5×10^8 m/s having a wavelength of 0.5 mm in a non-magnetic good conductor. Calculate the frequency of wave and conductivity of the medium.
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